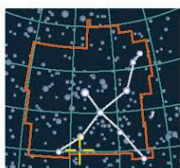


## SUPERNOVA REMNANT

## The Cygnus Loop



CYGNUS

**CATALOG NUMBER**  
NGC 6960/95

**DISTANCE FROM SUN**  
2,600 light-years

**MAGNITUDE** 11

the Veil Nebula, and because it is so large, the Cygnus Loop has been cataloged using many different reference numbers. The supernova remnant is some 80 light-years long and sprawls 3.5 degrees across the sky—about seven full moons across. It shines in the light generated by shock waves

## GLOWING FILAMENTS

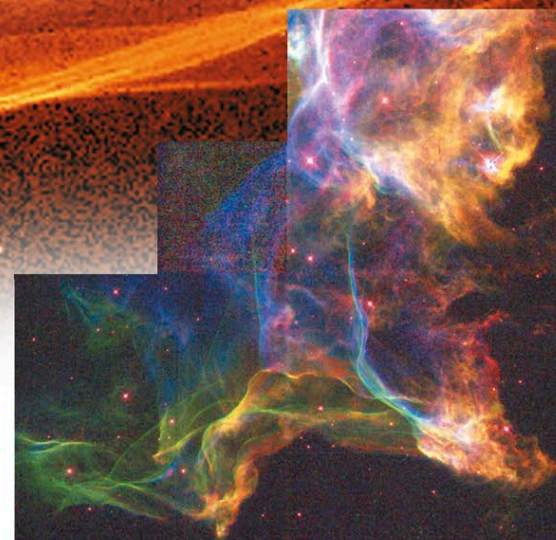
Filaments of shocked interstellar gas glow in the light emitted by excited hydrogen atoms. This side-on view shows a small portion of the Cygnus Loop moving upward at about 380,000 mph (612,000 kph).

The Cygnus Loop is the remnant of a dying star that blew itself up in a supernova. Estimates of how long ago in the star's lifetime this event occurred vary from 5,000 to 15,000 years. The most prominent parts of the nebula seen in visible light are often called

produced as stellar material from the supernova hits material in the interstellar medium. Observations of this stellar laboratory have revealed an inconsistent composition and structure of the interstellar medium, as well as that of the supernova remnant.

## COLORFUL GASES

This composite image of a section of the Cygnus Loop reveals the presence of different kinds of atoms excited by shock waves: oxygen (blue), sulfur (red), and hydrogen (green).



## SUPERNOVA REMNANT

## Vela Supernova



VELA

**CATALOG NUMBER**  
NGC 2736

**DISTANCE FROM SUN**  
6,000 light-years

**MAGNITUDE** 12

The Vela Supernova Remnant is the brightest object in the sky at gamma-ray wavelengths. It is estimated that the star that produced it exploded between 5,000 and 11,000 years previously, and that its final explosion would have rivaled the Moon as the brightest object in the night sky. The star that died has become a pulsar, a rapidly spinning neutron star, which rotates about 11 times each second. The Vela Pulsar is about 12 miles (19 km) in diameter and was only the second pulsar to be discovered optically, the optical flashes being observed in 1977. As with other pulsars, the rotation rate of the Vela Pulsar is gradually slowing down. Since 1967, it has suffered several brief glitches where its rotation rate has temporarily increased before continuing to slow.

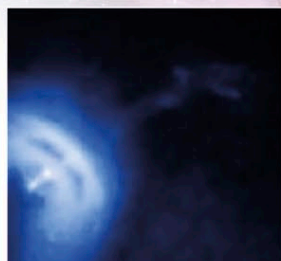
## DYNAMIC JET

A sequence of X-ray images from the Chandra satellite reveals changes in the jet of high-energy particles emitted by the Vela Pulsar. The jet remains tightly structured for more than half a light-year before losing coherence due to contact with surrounding interstellar medium.

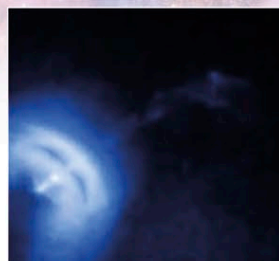


## EXPANDING SHELL

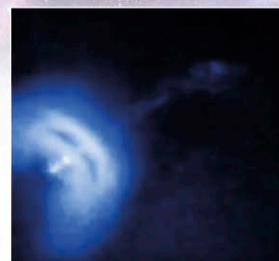
This optical photograph of the Vela Supernova Remnant reveals details of the 100-light-year-wide expanding gas bubble against more distant nebulosity.



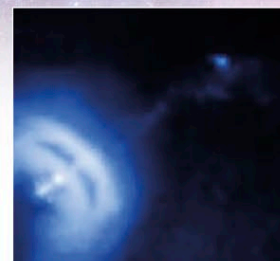
JUNE 2010



JULY 2010



AUGUST 2010



SEPTEMBER 2010